

Windows Active Directory & GPO Security Lab

Centralized Firewall Policy, Domain Management
and Client-Side Validation

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Project Overview

Centralized Windows security management

The lab built a small Active Directory environment in which a Windows Server acts as the domain controller and distributes a centrally defined Windows Firewall policy to two managed clients through Group Policy.

Implemented

- ▶ Windows Server 2022 with AD DS and DNS
- ▶ RRAS/NAT for client Internet access
- ▶ Two domain-joined Windows clients
- ▶ Domain-linked firewall GPO
- ▶ Client-side policy verification

Core security idea

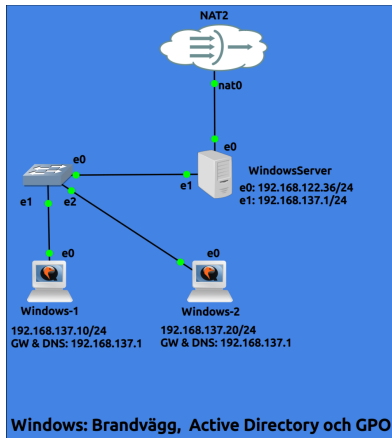
Define the policy once, distribute it centrally, and verify that the same control reaches every managed endpoint.

Out of scope

To keep the project focused, this presentation does not cover:

- ▶ Full Windows Firewall rule-set documentation
- ▶ Detailed AD DS and DNS configuration walkthroughs
- ▶ Every Group Policy setting available in the domain
- ▶ Production-scale monitoring, logging and policy design

Lab Architecture



Windows Server

External interface: 192.168.122.36/24

Internal interface: 192.168.137.1/24

Services:

- ▶ AD DS
- ▶ DNS
- ▶ RRAS / NAT
- ▶ Group Policy

Domain clients

Windows-1 (CLIENT-1)

192.168.137.10/24

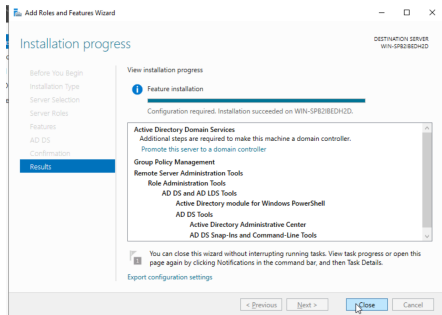
Windows-2 (CLIENT-2)

192.168.137.20/24

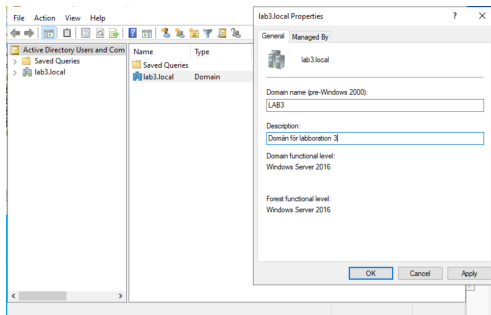
Gateway & DNS: 192.168.137.1

Active Directory Domain Services

AD DS installation



Domain controller configuration



What was implemented

Active Directory Domain Services was installed on Windows Server 2022 and the server was promoted to a domain controller for the `lab3.local` domain.

The domain controller became the central point for authentication, DNS and policy management.

RRAS NAT — Internet Access for Domain Clients

The screenshot shows the Windows Routing and Remote Access console. On the left, the tree view is expanded to 'WIN-SPB2IBEDH2D (local)' > 'Network Interfaces' > 'Ethernet 2'. The main pane displays the 'NAT' configuration for 'Ethernet 2', showing 28 total mappings and 829 inbound packets translated. On the right, the 'WIN-SPB2IBEDH2D - Network Address Translation Session Mapping Table' is open, displaying a table of active NAT sessions.

Protocol	Direction	Private address	Private port	Public Address
TCP	Outbound	192.168.137.10	57 799	192.168.122.36
TCP	Outbound	192.168.137.20	61 718	192.168.122.36
TCP	Outbound	192.168.137.20	61 727	192.168.122.36
TCP	Outbound	192.168.137.20	61 739	192.168.122.36
TCP	Outbound	192.168.137.10	58 259	192.168.122.36

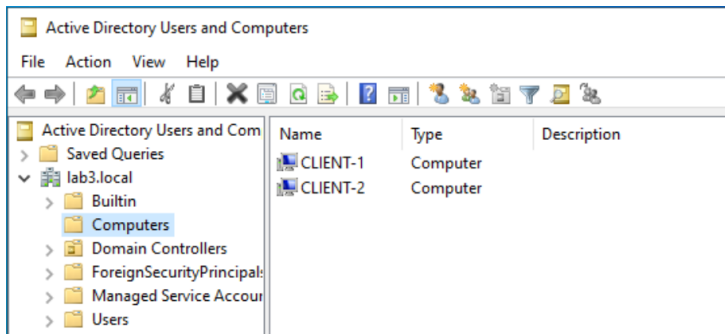
Two-interface design

The server used an internal/private interface for the domain network and an external/public interface for Internet access.

Evidence in the capture

RRAS shows active NAT mappings from internal client addresses to the server's external address, confirming translated outbound connectivity.

Domain-Joined Windows Clients

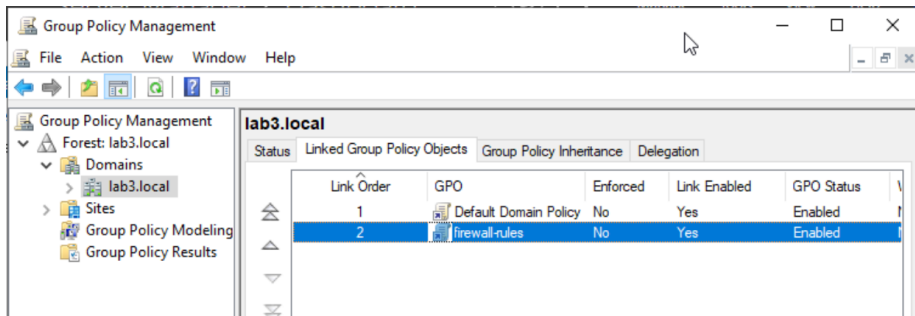


Managed endpoints

Active Directory Users and Computers shows both **CLIENT-1** and **CLIENT-2** registered as computer objects in `lab3.local`.

Once joined to the domain, the clients can receive centrally managed security settings instead of relying only on local configuration.

Centralized Firewall Policy with GPO



Policy design

A Group Policy Object named `firewall-rules` was linked to the `lab3.local` domain.

Security value

The same firewall configuration can be distributed from one management point to multiple Windows endpoints.

GPO Deployment and Verification

Force policy refresh

```
C:\Users\Administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.
```

Deployment step

gpupdate /force refreshed computer policy so the new GPO could be processed immediately during testing.

Verify the applied GPO

```
COMPUTER SETTINGS
-----
CN-CLIENT-1,CN-Computers,DC-lab3,DC-local
Last time Group Policy was applied: 2026-04-13 at 20:18:23
Group Policy was applied from: WIN-SPB2I8EOH2D.lab3.local
Group Policy slow link threshold: 500 kbps
Domain Name: LAB3
Domain Type: Windows 2008 or later

Applied Group Policy Objects
-----
firewall-rules
Default Domain Policy
```

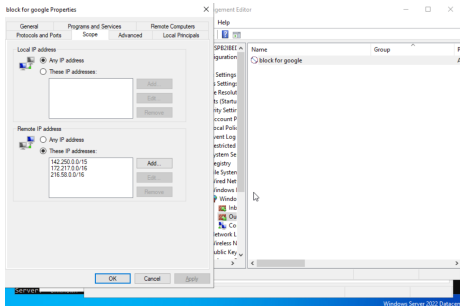
```
COMPUTER SETTINGS
-----
CN-CLIENT-2,CN-Computers,DC-lab3,DC-local
Last time Group Policy was applied: 2026-04-13 at 20:45:28
Group Policy was applied from: WIN-SPB2I8EDH2D.lab3.local
Group Policy slow link threshold: 500 kbps
Domain Name: LAB3
Domain Type: Windows 2008 or later

Applied Group Policy Objects
-----
firewall-rules
Default Domain Policy
```

Observed result

gpresult lists firewall-rules under Applied Group Policy Objects on both clients. This verifies delivery, not merely configuration on the domain controller.

Outbound Firewall Rule — Google Access Test



The GPO defined an outbound Windows Firewall rule containing several remote IP ranges used for the Google blocking experiment.

Purpose of the test

The goal was to prove centralized enforcement across the domain, not to present static IP lists as an ideal production web-filtering method.

Functional Test — Blocked Target, Allowed Control

```
C:\Users\Administrator>ping google.com

Pinging google.com [142.251.38.110] with 32 bytes of data:
General failure.
General failure.

Ping statistics for 142.251.38.110:
    Packets: Sent = 2, Received = 0, Lost = 2 (100% loss),
Control-C
^C
C:\Users\Administrator>ping aftonbladet.se

Pinging aftonbladet.se [35.71.185.87] with 32 bytes of data:
Reply from 35.71.185.87: bytes=32 time=16ms TTL=247
Reply from 35.71.185.87: bytes=32 time=44ms TTL=247
Reply from 35.71.185.87: bytes=32 time=50ms TTL=247

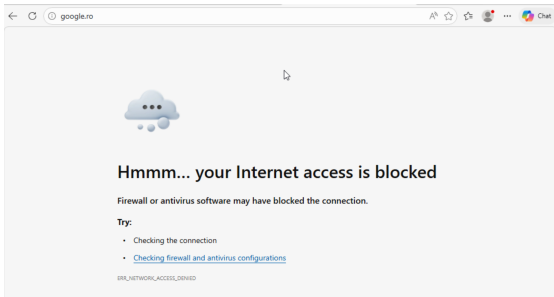
Ping statistics for 35.71.185.87:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 50ms, Average = 36ms
Control-C
```

Observed result

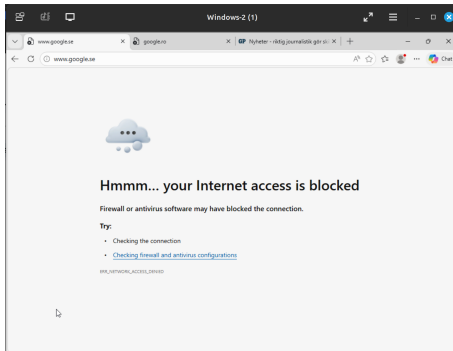
The client could not reach `google.com`, while a control test to an unrelated destination succeeded. This is stronger evidence than a failed Google test alone because it shows that general connectivity still worked.

Browser Validation — Same Result on Both Clients

CLIENT-1



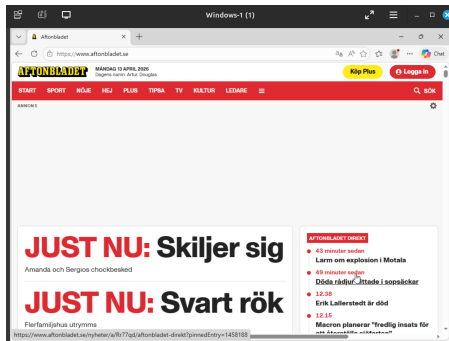
CLIENT-2



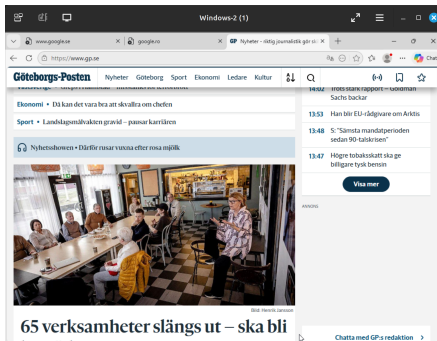
Both endpoints show the same firewall-related browser failure, supporting consistent policy enforcement across the managed clients.

Control Test — Other Websites Remained Reachable

CLIENT-1 — allowed website



CLIENT-2 — allowed website



Why the control test matters

Unrelated sites remained reachable, indicating that the clients still had Internet access and that the test rule did not simply break all outbound web traffic.

Limitation — IP-Based Service Blocking

What the lab revealed

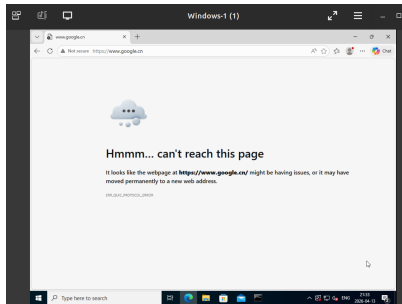
Large Internet services use many addresses and multiple protocols.

A static IP list therefore becomes difficult to maintain and may produce different failure behaviour across domains or connections.

Practical lesson

The GPO deployment mechanism worked well, but the specific IP-based filtering method is a limited form of service control.

Observed edge case



A different browser error appeared during one Google-domain test, showing that application behaviour can vary even when access still fails.

Why Centralized GPO Enforcement Matters

Consistency

One policy can be applied to multiple managed endpoints.

Scalability

Changes are distributed centrally instead of configured host by host.

Verification

Client tools confirm whether the intended policy actually arrived.

Main security lesson

The strongest result of the lab is the management model: Active Directory and Group Policy provide a centralized and verifiable mechanism for enforcing Windows endpoint security settings.

Troubleshooting and Lessons Learned

1. Delivery

Creating the GPO was only the first step.

Policy refresh and endpoint verification were required to prove deployment.

2. Connectivity

The clients needed domain DNS and Internet access at the same time.

RRAS NAT supported that requirement in the virtual lab.

3. Precision

Blocking a large service by IP range was less predictable than the original simple rule suggested.

Practical takeaway

Validate centrally managed controls from the endpoint's perspective.

A rule visible on the management server does not by itself prove that the client has received or enforced it.

Production Security Considerations

Lab approach

- ▶ One domain-linked firewall GPO
- ▶ Two managed clients
- ▶ Windows Server also providing RRAS/NAT
- ▶ Static remote-IP ranges for the test rule

Production improvements to consider

- ▶ OU-based and security-group policy scoping
- ▶ Dedicated network-edge/NAT infrastructure
- ▶ Centralized firewall and event logging
- ▶ Application- or domain-aware controls where appropriate

Important distinction

These are design considerations identified after the lab.
They are not being claimed as features implemented in the original exercise.

Project Summary

Implemented

- ▶ Windows Server 2022 with AD DS and DNS
- ▶ RRAS NAT for the internal clients
- ▶ CLIENT-1 and CLIENT-2 joined to lab3.local
- ▶ Domain-linked firewall-rules GPO

Validated

- ▶ Successful policy refresh
- ▶ gpresult evidence on both endpoints
- ▶ Consistent blocking behaviour on both clients
- ▶ Control traffic to unrelated sites remained functional

Result

The project demonstrated a complete management path:

Domain configuration

→

GPO deployment

→

client verification

→

functional testing.

It also showed the difference between a successful policy-deployment mechanism and the limitations of the filtering method used in the test.